

Mahesh C Talluri

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Professional Summary

Results-driven Data Engineer with 3+ years of hands-on experience designing and building scalable data pipelines, data models, and automation workflows across cloud environments. Proven expertise in **AWS (Glue, S3, Lambda, Redshift, EMR)** and **Databricks** for ETL, Spark-based processing, and data transformation at scale. Proficient in developing reusable frameworks, optimizing big data jobs, and ensuring data quality through test-driven development and validation techniques. Experienced in implementing **CI/CD pipelines**, collaborating with cross-functional teams, and analyzing complex datasets for root cause resolution and continuous data improvement. Strong foundation in the **software development life cycle (SDLC)**, scripting, and modern data warehousing practices.

EDUCATION

New Jersey Institute of Technology

Master of Science - Computer Science, GPA: 3.75

Newark, NJ

May 2023

Visvesvaraya National Institute of Technology

Bachelor of Technology - Computer Science

Nagpur, India

May 2021

SKILLS

- **Programming Languages:** Python, SQL, R, Java, Shell
- **Cloud Platforms:** AWS (Glue, Lambda, S3, Redshift, EMR, SageMaker), Databricks
- **ETL & Pipelines:** ADF Pipelines, SQL-based transformations, Python ETL scripts, Azure Functions, Apache Airflow.
- **Big Data & Streaming:** Spark(SQL, Streaming, MLlib), Hadoop, Hive, Kafka, HDFS
- **ML & Data Science:** Classification, Regression, Ranking, Forecasting, Recommender Systems, A/B Testing
- **Visualization:** Power BI, Tableau, Amazon QuickSight, Streamlit(familiar)
- **Databases:** Snowflake, Redshift, PostgreSQL, MySQL, NOSQL
- **Data Management:** Data Modeling, schema design, model interpretation & Test Driven Development
- **Other Tools:** Git, Jupyter, CloudWatch, JSON, Parquet, ORC, XML, Avro

PROFESSIONAL EXPERIENCE

Massachusetts Financial Services | Data Engineer

Jan 2023 - Present

- Designed and maintained reusable **ETL** pipelines using **Python, SQL**, and **AWS Glue** to support real-time and batch processing needs.
- Migrated and restructured legacy data architecture into cloud-native solutions using **Redshift, S3**, and **Lambda**.
- Developed and deployed **CI/CD** pipelines using **Git** and **DevOps** tools to enable automated testing and deployment.
- Implemented **Spark**-based data transformation on **AWS EMR, Databricks** for large-scale customer and transaction data.
- Created scalable data models to support analytics and reporting across finance and marketing teams.
- Built test plans, root cause analysis documentation, and reusable frameworks to improve data reliability.

Reliance Jio | Data Analyst

Jan 2021 - Jan 2022

- Extracted, transformed, and validated telecom and financial datasets using Hive, SQL Server, and Oracle.
- Automated ETL processes with Python and shell scripts to improve SLA adherence and reduce manual tasks.
- Collaborated with business teams to develop views and dashboards in Tableau for customer insights and churn prediction.
- Conducted unit testing and QA checks for all transformation jobs to ensure completeness and accuracy.

PROJECTS

ML Vulnerability Detection

Tool(Vinci) (ML4Cyber)

(Python, sci-kit-learn, TensorFlow, Machine Learning, Deep Learning)

- Designed a PyQt5-based interface to enhance data import and labeling for efficient code analysis, integrating PMD, PyLint, and ESLint, while adding group labeling, large dataset processing, and file conversion features.
- Developed a machine learning tool to predict software vulnerabilities using CWE-labeled data. Achieved a 15% improvement in prediction accuracy by employing data balancing techniques and utilizing Python's scikit-learn and TensorFlow.

Student Records Analytics Pipeline

(Big Data, AWS Glue, EMR, Redshift, QuickSight)

- Built a scalable data pipeline using AWS Glue and Python, processing over 100,000 student records by employing distributed processing and custom data partitioning, leading to a 30% improvement in processing efficiency.
- Automated ETL workflows with AWS Lambda by optimizing memory allocation and implementing parallel data transformations, reducing data processing time by 30%.
- Optimized data storage and retrieval by implementing lifecycle policies in Amazon S3 and tuning Redshift queries, leading to a 40% reduction in query times and enhanced data accessibility for analytics.

CERTIFICATIONS

- AWS Certified Solution Architect Associate (SAA-C03)
- Data Engineering using AWS Data Analytics